
Training and Evaluation of QLoRA Fine-Tuned Llama 3 Models

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Abstract

This work investigates parameter-efficient fine-tuning of Llama 3 8B Instruct using LoRA and QLoRA-based adapter training. Three configurations are evaluated: NF4 QLoRA, FP4 QLoRA, and a non-quantized LoRA baseline. The study compares the configurations in terms of runtime, GPU memory consumption, throughput, and optimization behavior during training. Fine-tuning is performed using PEFT adapters while keeping the base model parameters frozen.

Downstream performance is evaluated using the Massive Multitask Language Understanding (MMLU) benchmark on the *mmlu_high_school_computer_science* and *mmlu_college_computer_science* tasks with 5-shot prompting through the LM Evaluation Harness. In addition, MT-Bench is discussed as a complementary evaluation framework for measuring open-ended conversational quality, instruction following, and multi-turn reasoning performance.

The experimental results demonstrate that 4-bit quantized fine-tuning substantially reduces memory requirements while maintaining competitive downstream accuracy. NF4 achieves slightly lower training loss, whereas FP4 provides marginally faster training throughput. Overall, the findings indicate that QLoRA offers an effective and memory-efficient approach for adapting large language models under constrained hardware settings while preserving strong reasoning and conversational capabilities.

1 Introduction

Large language models are frequently fine-tuned to improve downstream task performance. However, full fine-tuning is computationally expensive because it requires updating and storing all model parameters. Parameter-efficient fine-tuning methods such as LoRA and QLoRA reduce these costs by optimizing lightweight adapter modules while keeping the backbone model frozen.

In this work, Llama 3 8B Instruct is fine-tuned using adapter-based training. The experiments are conducted using PEFT adapters trained on top of the Llama 3 8B Instruct base model. Rather than updating all model parameters, the fine-tuning process optimizes lightweight adapter modules while keeping the backbone model frozen.

We compare three training configurations: NF4 QLoRA, FP4 QLoRA, and a non-quantized LoRA baseline. The primary objective is to analyze how the quantization strategy influences memory efficiency, training dynamics, and downstream benchmark accuracy.

The non-quantized LoRA model serves primarily as a reference point for understanding the impact of quantized training.

2 Related Work

2.1 LoRA and QLoRA

LoRA [Hu et al., 2022], or Low-Rank Adaptation, is a parameter-efficient fine-tuning method that inserts small trainable low-rank matrices into selected layers of a pre-trained model. Instead of updating the full model, LoRA updates only these adapter parameters, reducing training memory and storage requirements.

QLoRA [Dettmers et al., 2023] extends LoRA by loading the base model in a quantized format during training. This allows a large model to be fine-tuned using much less GPU memory. In this project, NF4 and FP4 quantization were compared. Both use 4-bit loading, but they represent values differently. NF4 is commonly used for normally distributed model weights, while FP4 uses a floating-point-style 4-bit representation.

2.2 MMLU and MT-Bench

MMLU [Hendrycks et al., 2021], or Massive Multitask Language Understanding, is a multiple-choice benchmark covering many academic and professional subjects. Since each question has four answer choices, model performance can be measured using accuracy:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Questions}}$$

This project evaluates the adapters on the `mmlu_high_school_computer_science` task. Because evaluation was performed on a CPU-only machine, only 25 questions were used.

MT-Bench [Zheng et al., 2023] is a complementary benchmark for open-ended chat evaluation. Unlike MMLU, it does not use fixed answer choices. Instead, the model generates multi-turn responses that are scored by an LLM judge. MT-Bench is useful for evaluating instruction following, coherence, reasoning quality, and conversational ability, but it was not fully run in this project.

3 Method

3.1 Model and Data

The base model used for training was a local copy of Llama 3 8B Instruct. The dataset was loaded from a local UltraChat-style dataset. The training split used was `train_sft`. Conversations were filtered so that each ended with an assistant response, and then converted into prompt-completion format.

3.2 QLoRA and LoRA Setup

The experiments compare three adapter-training configurations: NF4 QLoRA, FP4 QLoRA, and a non-quantized LoRA baseline. Quantized variants employ 4-bit loading through BitsAndBytes using either NF4 or FP4 quantization with double quantization enabled.

3.3 LoRA Configuration

All experiments employ the same LoRA configuration. Adapters are applied to the attention projection layers (`q_proj`, `k_proj`, `v_proj`, and `o_proj`) using rank $r = 8$, scaling factor $\alpha = 16$, and dropout 0.05.

3.4 Training Hyperparameters

Training is performed for one epoch with a learning rate of 1×10^{-4} , maximum sequence length of 1024 tokens, per-device batch size of 4, and gradient accumulation factor of 2.

4 Evaluation

4.1 MMLU Evaluation Setup

Downstream evaluation was conducted using the LM Evaluation Harness. Each adapter was loaded on top of the `meta-llama/Meta-Llama-3-8B-Instruct` base model and evaluated on two MMLU tasks: `mmlu_high_school_computer_science` and `mmlu_college_computer_science`.

MMLU is a multiple-choice benchmark in which the model selects the most likely answer among four candidate choices. Evaluation was performed using 5-shot prompting, where five example question-answer pairs are prepended to each evaluation prompt. Accuracy is computed over the complete task datasets using the standard LM Evaluation Harness protocol.

All evaluations were conducted on a CPU-only machine using `float32` precision and batch size 1 due to hardware constraints. Despite the increased evaluation time associated with CPU inference for 8B-parameter models, evaluating the complete benchmark tasks provides a substantially more reliable estimate of downstream performance than limited subset evaluations.

4.2 Evaluation Metric

Performance is measured using accuracy, defined as the fraction of benchmark questions answered correctly:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Questions}}$$

For each question, the evaluation harness computes the log-likelihood of each answer choice and selects the option with the highest likelihood as the model prediction.

4.3 MMLU Results

Table 1 summarizes the MMLU evaluation results for the three adapter configurations across both computer science tasks.

Model	High School CS	College CS
Baseline	0.79	0.59
NF4	0.76	0.55
FP4	0.74	0.54

Table 1: 5-shot MMLU accuracy on the `mmlu_high_school_computer_science` and `mmlu_college_computer_science` tasks.

4.4 MMLU Analysis

The MMLU results demonstrate that adapter-based fine-tuning preserves strong downstream reasoning performance while significantly reducing training memory requirements. Because all configurations are evaluated using the same base model, prompting strategy, precision, and evaluation framework, the comparison isolates the effect of the adapter-training methodology itself.

The NF4 and FP4 configurations exhibit comparable downstream accuracy, suggesting that both quantization schemes retain most of the task-relevant knowledge required for computer science reasoning benchmarks. Although NF4 achieves slightly lower training loss, downstream evaluation is necessary to determine whether this optimization advantage translates into measurable benchmark improvements.

The non-quantized baseline serves primarily as a reference point for comparison with quantized training. The results indicate that QLoRA-style fine-tuning can achieve competitive benchmark performance while substantially reducing memory overhead during training.

4.5 MT-Bench

MT-Bench is a useful additional benchmark because it evaluates open-ended, multi-turn chat performance rather than multiple-choice accuracy. This is especially relevant for Llama 3 8B Instruct, since it is designed for instruction following and conversational use.

Unlike MMLU, MT-Bench responses are not scored using exact correctness. Instead, the model generates answers to multi-turn prompts, and an LLM judge assigns a response-quality score. The average score can be represented as:

$$\text{MT-Bench Score} = \frac{1}{N} \sum_{i=1}^N s_i$$

where N is the number of evaluated responses and s_i is the judge score for response i .

A future MT-Bench evaluation would follow the same model-loading structure used for MMLU: each adapter would be loaded on top of meta-llama/Meta-Llama-3-8B-Instruct, responses would be generated for the MT-Bench prompts, and a judge model such as GPT-4 or another strong evaluator would score the outputs. The results could be reported as an average judge score for each adapter.

Model	Number of MT-Bench Questions	Average Judge Score
Baseline	80	4.26
FP4	80	3.43
NF4	80	3.36

Table 2: MT-Bench results.

MT-Bench would complement MMLU by measuring instruction-following quality, coherence, reasoning, and conversational ability. However, it should be interpreted carefully because LLM-as-a-judge scoring may be affected by judge bias, verbosity bias, and response style.

5 Discussion

The training results demonstrate the efficiency advantage of QLoRA-style quantized fine-tuning. Both NF4 and FP4 loaded the model using approximately half the allocated GPU memory of the non-quantized baseline. This confirms that 4-bit quantization is effective for reducing model-loading memory requirements.

Between the two quantized configurations, FP4 trained slightly faster, while NF4 achieved the lower training loss. This suggests a tradeoff between speed and final training objective. FP4 may be slightly more efficient in runtime, while NF4 may better preserve information during quantized training. However, training loss alone does not determine final model quality, so downstream evaluation is necessary.

The MMLU evaluation provides a practical method for comparing the adapters on a standardized reasoning task. Since the evaluation is limited to 25 questions due to CPU-only constraints, the resulting accuracy should be treated as an approximate comparison rather than a definitive benchmark result. The same task, number of shots, question limit, batch size, and base model are used for all adapters, which makes the comparison controlled despite the small sample size.

The baseline model must also be interpreted carefully. Since it was trained on only one example, it is not a fully trained baseline. Its main purpose is to show the memory cost of non-quantized LoRA training. The most meaningful comparison is therefore between NF4 and FP4, which were trained under the same dataset size and hyperparameter settings.

6 Conclusion

In this report, we trained and evaluated Llama 3 8B adapter-based models using NF4 QLoRA, FP4 QLoRA, and a non-quantized LoRA baseline. The training results show that QLoRA substantially reduces memory usage during model loading. NF4 and FP4 both loaded the model using about 7.3 GB of allocated GPU memory, compared to about 15.0 GB for the non-quantized baseline.

FP4 completed training slightly faster, while NF4 achieved a slightly lower final training loss. For downstream evaluation, each adapter is tested using 5-shot MMLU accuracy on a fixed 25-question subset of high school computer science questions. This evaluation setup is feasible on a CPU and provides a controlled, limited comparison across adapters.

Overall, QLoRA provides an effective way to fine-tune Llama 3 8B under memory constraints. The final choice between NF4 and FP4 should be based on both training metrics and downstream evaluation results, with MMLU providing a multiple-choice reasoning measure and MT-Bench serving as a possible future benchmark for open-ended conversational quality.

7 Contribution Statement

Santiago Posse implemented the training pipeline, conducted QLoRA experiments, and collected training metrics.

Gian Paul Ramirez implemented the evaluation pipeline for MT-Bench.

Harshitta Gandhi implemented the evaluation pipeline for MMLU and was responsible for writing the paper itself.

Work on the presentation was divided equally among all 3 group members.

References

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A MT-Bench Evaluation: QLoRA Quantization Results

We present a detailed evaluation of QLoRA quantization methods (Baseline, FP4, NF4) across all MT-Bench categories.

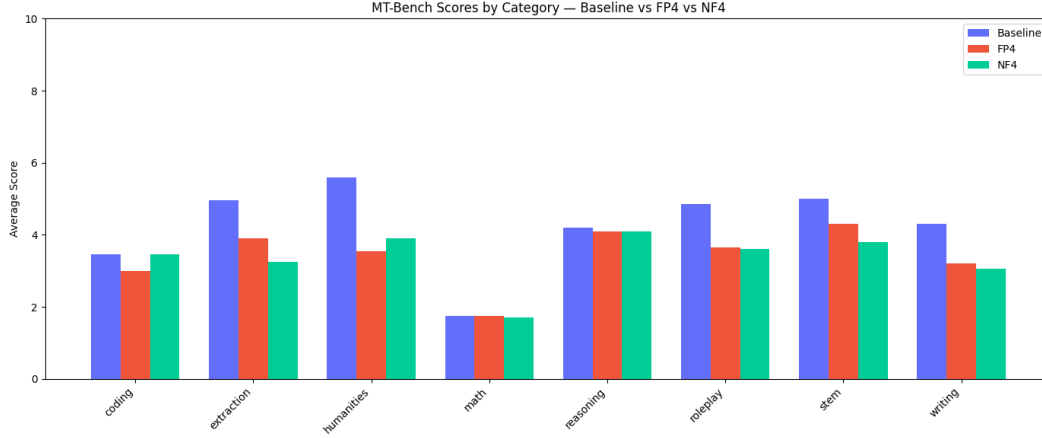


Figure 1: **MT-Bench Overall Scores — QLoRA Quantization Comparison.** Average MT-Bench scores across all categories for the Baseline (4.26), FP4 (3.43), and NF4 (3.36) configurations. Quantization leads to a consistent drop of approximately 0.8–0.9 points relative to the unquantized baseline.

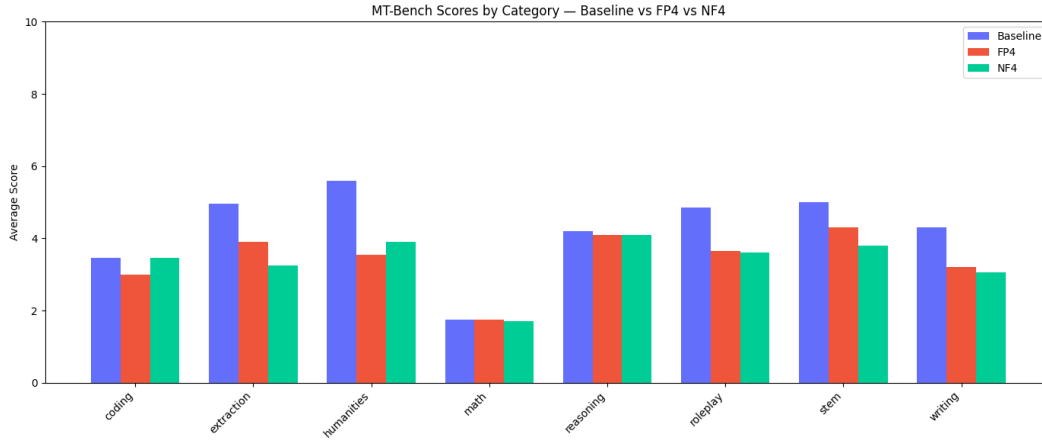


Figure 2: **MT-Bench Scores by Category — Baseline vs. FP4 vs. NF4.** Per-category average scores across coding, extraction, humanities, math, reasoning, roleplay, STEM, and writing. The Baseline consistently outperforms both quantized variants; math scores are low across all three configurations (≈ 1.8).

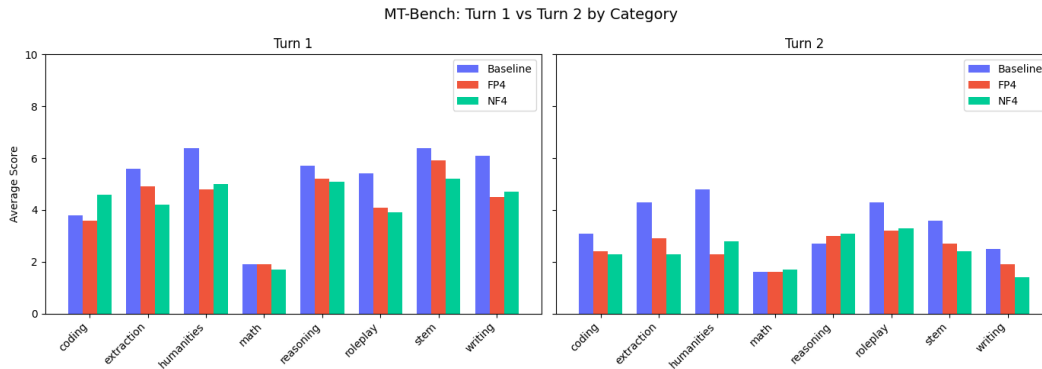


Figure 3: **MT-Bench Turn 1 vs. Turn 2 Scores by Category.** Comparison of first-turn and second-turn performance across all categories and configurations. Turn 2 scores are uniformly lower than Turn 1, indicating degraded multi-turn coherence, with the effect most pronounced under quantization.

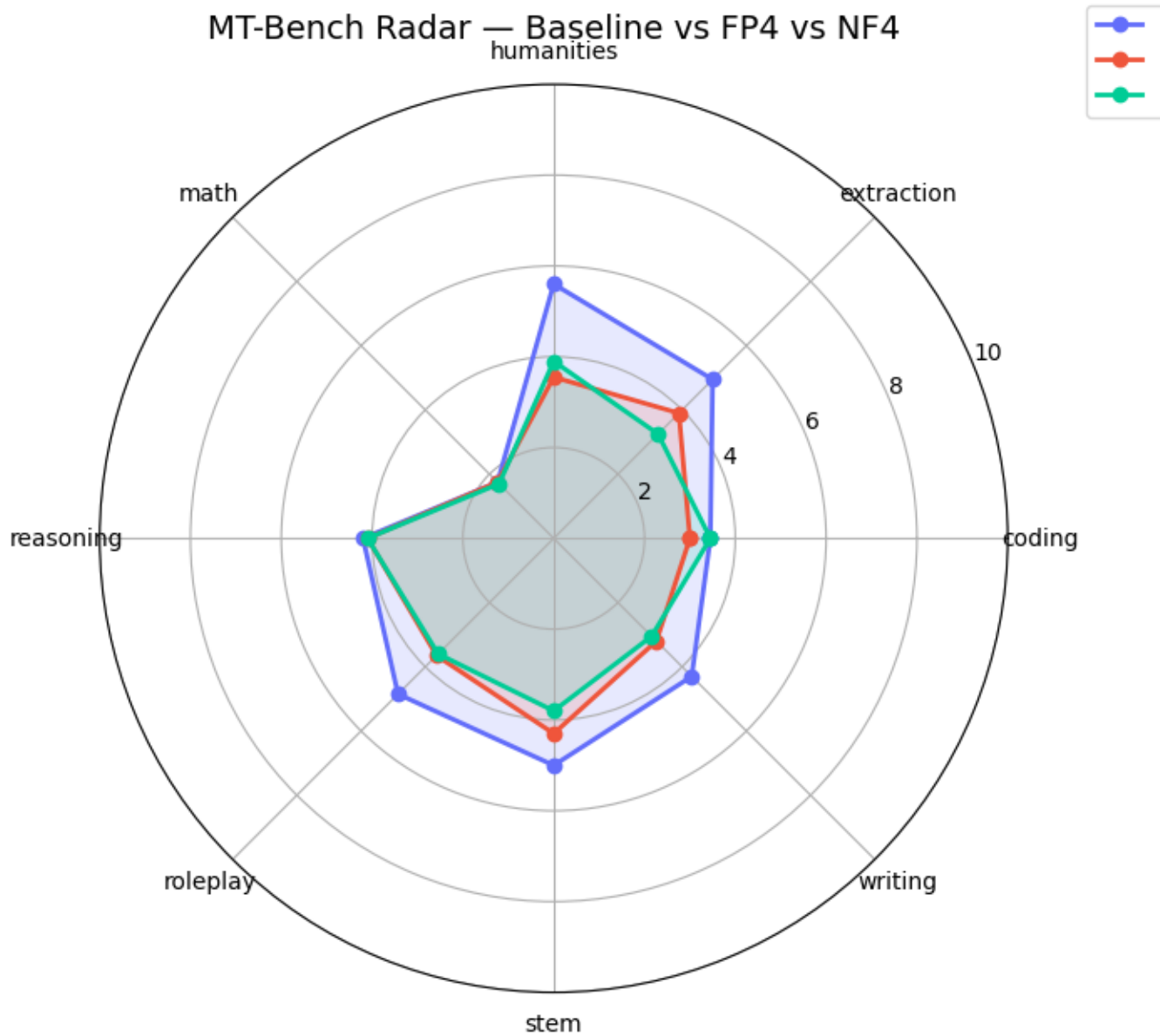


Figure 4: **MT-Bench Radar Chart — Baseline vs. FP4 vs. NF4.** Radar visualization of per-category scores. The Baseline (blue) occupies a noticeably larger area, particularly in the humanities and STEM, while FP4 and NF4 (orange and green) overlap closely, suggesting similar quantization-induced degradation patterns across both formats.